Appendix B

Linear bound propagation with split neuron constraints

B. *Proof of Theorem 2*

Theorem 2: Given a trained multi-input-single-output neural network $f: \mathbb{R}^{d_0} \to \mathbb{R}$, with weights $\boldsymbol{W}^{(i)}$ and biases $\boldsymbol{b}^{(i)}$ with input bound $\boldsymbol{x} \in \mathcal{B}(\boldsymbol{x}_{ref}, \boldsymbol{\epsilon}^+, \boldsymbol{\epsilon}^-)$ and split neuron constraints $z \in \mathcal{U}$, we have:

$$\min_{\boldsymbol{x}\in\mathcal{B}, z\in\mathcal{U}} f(\boldsymbol{x}) \geq \max_{\boldsymbol{\beta}\geq 0} \min_{\boldsymbol{x}\in\mathcal{B}} (\boldsymbol{a} + \boldsymbol{P}\boldsymbol{\beta})^T \boldsymbol{x} + \boldsymbol{q}^{\mathrm{T}}\boldsymbol{\beta} + c$$

where $\boldsymbol{a} \in \mathbb{R}^{d_0}, \boldsymbol{P} \in \mathbb{R}^{d_0 \times (\sum_{i=1}^{L-1} d_i)} := \left[\boldsymbol{P_1}^{\mathrm{T}}\ \boldsymbol{P_2}^{\mathrm{T}} \dots\ \boldsymbol{P_{L-1}}^{\mathrm{T}}\right], \boldsymbol{q} \in \mathbb{R}^{\sum_{i=1}^{L-1} d_i} := [\boldsymbol{q_1}^{\mathrm{T}}\ \dots\ \boldsymbol{q_{L-1}}^{\mathrm{T}}]^{\mathrm{T}}$ and $c \in \mathbb{R}$ are functions of $\boldsymbol{W}^{(i)}, \boldsymbol{b}^{(i)}, \boldsymbol{l}^{(i)}, \boldsymbol{u}^{(i)}$.

Detailed formulations for $\boldsymbol{a}, \boldsymbol{P}, \boldsymbol{q}$ and c are given below,

$$\boldsymbol{\Psi}(i,i) = \boldsymbol{I}, \boldsymbol{\Psi}(k+1,i) = \boldsymbol{W}^{(k+1)} \underline{\boldsymbol{D}}^{(k)} \boldsymbol{\Psi}(k,i)$$

$$\boldsymbol{a} = \left[\boldsymbol{\Psi}(L,1)\boldsymbol{W}^{(1)}\right]^{\mathrm{T}}$$

$$\boldsymbol{P}_i = \boldsymbol{K}^{(i)} \boldsymbol{\Psi}(i,1) \boldsymbol{W}^{(1)} \quad \forall i \in [1, L-1]$$

$$\boldsymbol{q}_i = \sum_{k=1}^{i} \boldsymbol{K}^{(i)} \boldsymbol{\Psi}(i,k) \boldsymbol{b}^{(k)} + \sum_{k=2}^{i} \boldsymbol{K}^{(i)} \boldsymbol{\Psi}(i,k) \boldsymbol{W}^{(k)} \underline{\boldsymbol{D}}^{(k-1)} \mathrm{diag}(\underline{\boldsymbol{E}}^{(k-1)}) \quad \forall i \in [1, L-1]$$

$$c = \sum_{i=1}^{L} \boldsymbol{\Psi}(L,i) \boldsymbol{b}^{(i)} + \sum_{i=2}^{L} \boldsymbol{\Psi}(L,i) \boldsymbol{W}^{(i)} \underline{\boldsymbol{D}}^{(i-1)} \mathrm{diag}(\underline{\boldsymbol{E}}^{(i-1)})$$

Diagonal matrices $\underline{\boldsymbol{D}}^{(i)}$ and $\underline{\boldsymbol{E}}^{(i)}$ are determined by the relaxation of ReLU neurons, and $\boldsymbol{A}^{(i)} \in \mathbb{R}^{1 \times d_i}$ represents the linear relationship between $f(\boldsymbol{x})$ and $\hat{\boldsymbol{z}}^{(i)}$. $\underline{\boldsymbol{D}}^{(i)}$ and $\underline{\boldsymbol{E}}^{(i)}$ depend on $\boldsymbol{A}^{(i)}$ such that:

$$\underline{D}_{j,j}^{(i)} = \begin{cases} 1 & if\ j \in \mathcal{U}^{+(i)} \\ 0 & if\ j \in \mathcal{U}^{-(i)} \\ \underline{\theta}_j & if\ j \notin \mathcal{U}^{+(i)} \cup \mathcal{U}^{-(i)}\ and\ A_j^{(i)} \geq 0 \\ \overline{\theta}_j & if\ j \notin \mathcal{U}^{+(i)} \cup \mathcal{U}^{-(i)}\ and\ A_j^{(i)} < 0 \end{cases} \qquad \forall j \in [1, d_i]$$

$$\underline{E}_{j,j}^{(i)} = \begin{cases} 0 & if\ j \in \mathcal{U}^{+(i)} \cup \mathcal{U}^{-(i)} \\ \underline{\delta}_j & if\ j \notin \mathcal{U}^{+(i)} \cup \mathcal{U}^{-(i)}\ and\ A_j^{(i)} \geq 0 \\ \overline{\delta}_j & if\ j \notin \mathcal{U}^{+(i)} \cup \mathcal{U}^{-(i)}\ and\ A_j^{(i)} < 0 \end{cases} \qquad \forall j \in [1, d_i]$$

$$\boldsymbol{A}^{(i)} = \begin{cases} \boldsymbol{W}^{(L)} & if\ i = L-1 \\ (\boldsymbol{A}^{(i+1)}\underline{\boldsymbol{D}}^{(i+1)} + \boldsymbol{\beta}^{(i+1)^{\mathrm{T}}}\boldsymbol{K}^{(i+1)})\boldsymbol{W}^{(i+1)} & if\ 0 \leq i \leq L-2 \end{cases}$$

Proof. Following the work in [B1], we prove this theorem by induction. Starting with the assumption that we know the bounds for layer $\hat{\boldsymbol{z}}^{(m)}$, we derive bounds for $\hat{\boldsymbol{z}}^{(m-1)}$ and continue this process until we reach $m = 0$. By definition, $\hat{\boldsymbol{z}}^{(0)} = \boldsymbol{x}$**.** We begin by defining a set of matrices and vectors $\boldsymbol{a}^{(m)}$, $\boldsymbol{P}^{(m)}$, $\boldsymbol{q}^{(m)}$, $c^{(m)}$, where $\boldsymbol{P}^{(m)} \in \mathbb{R}^{d_m \times (\sum_{i=m+1}^{L-1} d_i)}$ is a matrix containing blocks $\boldsymbol{P} \coloneqq \left[\boldsymbol{P}_{m+1}^{(m)^{\mathrm{T}}} \dots \boldsymbol{P}_{L-1}^{(m)^{\mathrm{T}}}\right]$, $\boldsymbol{q} \in \mathbb{R}^{\sum_{i=m+1}^{L-1} d_i}$ is a vector $\boldsymbol{q} \coloneqq \left[\boldsymbol{q}_{m+1}^{(m)^{\mathrm{T}}} \dots \boldsymbol{q}_{L-1}^{(m)^{\mathrm{T}}}\right]^{\mathrm{T}}$, and each term is defined as:

$$\boldsymbol{a}^{(m)} = \left[\Psi(L, m+1)\boldsymbol{W}^{(m+1)}\right]^{\mathrm{T}} \quad \in \mathbb{R}^{d_m} \tag{B1}$$

$$\boldsymbol{P}_i^{(m)} = \boldsymbol{K}^{(i)}\Psi(i, m+1)\boldsymbol{W}^{(m+1)} \in \mathbb{R}^{d_i \times d_m}, \qquad \forall i \in [m+1, L-1] \tag{B2}$$

$$\boldsymbol{q}_i^{(m)} = \sum_{k=m+1}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{b}^{(k)} + \sum_{k=m+2}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{W}^{(k)}\underline{\boldsymbol{D}}^{(k-1)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(k-1)}) \in \mathbb{R}^{d_i}$$

$$\forall i \in [m+1, L-1] \tag{B3}$$

$$c^{(m)} = \sum_{i=m+1}^{L} \Psi(L,i)\boldsymbol{b}^{(i)} + \sum_{i=m+2}^{L} \Psi(L,i)\boldsymbol{W}^{(i)}\underline{\boldsymbol{D}}^{(i-1)}\mathrm{diag}\left(\underline{\boldsymbol{E}}^{(i-1)}\right) \tag{B4}$$

and we claim that

$$\min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} f(x) \geq \max_{\widetilde{\boldsymbol{\beta}}^{(m+1)}\geq 0} \quad \min_{\boldsymbol{x}\in\mathcal{B},\widetilde{\boldsymbol{z}}^{(m)}\in\widetilde{\mathcal{U}}^{(m)}} \left(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right)^{\mathrm{T}}\hat{\boldsymbol{z}}^{(m)} + \boldsymbol{q}^{(m)^{\mathrm{T}}}\widetilde{\boldsymbol{\beta}}^{(m+1)}$$

$$+c^{(m)} \tag{B5}$$

where $\widetilde{\boldsymbol{\beta}}^{(m+1)} \coloneqq \left[\boldsymbol{\beta}^{(m+1)^{\mathrm{T}}} \ldots \boldsymbol{\beta}^{(L-1)^{\mathrm{T}}}\right]^{\mathrm{T}}$ concatenating all $\boldsymbol{\beta}^{(i)}$ variables up to layer $m+1$.

For the base $m = L-1$, we simply have,

$$\min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} f(x) = \min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} \boldsymbol{W}^{(L)}\hat{\boldsymbol{z}}^{(L-1)} + b^{(L)} \tag{B6}$$

No maximization is needed for this case since we have not yet reached the ReLU function in the last hidden layer. As such, there is no need to introduce Lagrange multipliers (i.e., β) at this stage. From (B1) and (B4) we have $\boldsymbol{a}^{(m)} = \left[\Psi(L,L)\boldsymbol{W}^{(L)}\right]^{\mathrm{T}} = \boldsymbol{W}^{(L)^{\mathrm{T}}}, c^{(m)} = \sum_{i=L}^{L}\Psi(L,i)\boldsymbol{b}^{(i)} = b^{(L)}$. All other terms are zero because no split multiplier exists at the output layer.

In Section III.C of the main paper, the intuition behind the proof was illustrated by demonstrating the backward propagation from the last hidden layer toward the preceding hidden layer. In particular, the propagation from $\hat{\boldsymbol{z}}^{(L-1)}$ to $\hat{\boldsymbol{z}}^{(L-2)}$ is presented in (24)–(26) of the main paper. We now generalize this process and prove the induction from $\hat{\boldsymbol{z}}^{(m)}$ to $\hat{\boldsymbol{z}}^{(m-1)}$.

We introduce a Lemma as follows,

Lemma B1: Given two vectors $w, v \in \mathbb{R}^d$, we have,

$$\boldsymbol{w}^{\mathrm{T}}\mathrm{ReLU}(\boldsymbol{v}) \geq \boldsymbol{w}^{\mathrm{T}}\underline{\boldsymbol{D}}\boldsymbol{v} + \boldsymbol{b}' \tag{B7}$$

where $\underline{\boldsymbol{D}}$ is a diagonal matrix containing the slopes of the linear bounds for a bounded neuron given as follows,

$$\underline{D}_{j,j} = \begin{cases} \underline{\theta}_j & if\ w_j \geq 0 \\ \overline{\theta}_j & if\ w_j < 0 \end{cases} \qquad \forall j \in [1, d] \tag{B8}$$

and $\boldsymbol{b}' = \boldsymbol{w}^{\mathrm{T}}\underline{\boldsymbol{D}}\mathrm{diag}(\underline{\boldsymbol{E}})$ where $\underline{\boldsymbol{E}}$ is a diagonal matrix containing the shift parameters of the linear bounds for a relaxed neuron as follows,

$$\underline{E}_{j,j} = \begin{cases} \underline{\delta}_j & if\ w_j \geq 0 \\ \overline{\delta}_j & if\ w_j < 0 \end{cases} \qquad \forall j \in [1, d] \tag{B9}$$

For a split neuron, the corresponding ReLU function is treated as linear within the selected subdomain. Thus, $D_{j,j} = 1$ and $E_{j,j} = 0$ for a positive split, whereas $D_{j,j} = 0$ and $E_{j,j} = 0$ for a negative split, consistent with (29.7) and (29.8) of the main paper.

Note that w_j are the elements of the equivalent weight matrix $\widetilde{\boldsymbol{W}}$ obtained through backward propagation of the linear bounds, consistent with the construction used in (A15). The proof for (B7) is as follows.

We know that $\boldsymbol{w}^{\mathrm{T}}\mathrm{ReLU}(\boldsymbol{v}) = \sum_j w_j \mathrm{ReLU}(v_j)$. For each term in this summation, we take the lower bound of $\mathrm{ReLU}(\boldsymbol{v}_j)$ if w_j is positive and the upper bound of $\mathrm{ReLU}(v_j)$ if w_j is negative (Reflected in the definitions of $\underline{\boldsymbol{D}}$ and $\underline{\boldsymbol{E}}$). This conservative choice allows us to always obtain a lower bound for a ReLU function:

$$\begin{aligned} \sum_j w_j \mathrm{ReLU}(v_j) &\geq \sum_j w_j \underline{D}_{j,j}\left(v_j + \underline{E}_{j,j}\right) = \boldsymbol{w}^{\mathrm{T}}\underline{\boldsymbol{D}}\boldsymbol{v} + \boldsymbol{w}^{\mathrm{T}}\underline{\boldsymbol{D}}\mathrm{diag}(\underline{\boldsymbol{E}}) \\ &= \boldsymbol{w}^{\mathrm{T}}\underline{\boldsymbol{D}}\boldsymbol{v} + \boldsymbol{b}' \end{aligned} \tag{B10}$$

Now we continue the proof for Theorem 2 by showing the induction from $\hat{\boldsymbol{z}}^{(m)}$ to $\hat{\boldsymbol{z}}^{(m-1)}$. Starting from (B5), since $\hat{\boldsymbol{z}}^{(m)} = \mathrm{ReLU}(\boldsymbol{z}^{(m)})$, we apply Lemma B1 by setting $\boldsymbol{w} =$

$\left[\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right] \coloneqq \boldsymbol{A}^{(m)^{\boldsymbol{T}}}$. It is easy to show that $\boldsymbol{A}^{(m)}$ can also be equivalently and recursively defined in (29.9) of the main paper. Based on Lemma B1, we have $\underline{\boldsymbol{D}}^{(m)}$ and $\underline{\boldsymbol{E}}^{(m)}$ defined as in (29.7) and (29.8) of the main paper, and hence (B5) becomes:

$$\min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} f(\boldsymbol{x}) \geq \max_{\widetilde{\boldsymbol{\beta}}^{(m+1)}\geq 0} \quad \min_{\boldsymbol{x}\in\mathcal{B},\tilde{\boldsymbol{z}}^{(m)}\in\tilde{\mathcal{U}}^{(m)}} \left(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right)^{\mathrm{T}} \underline{\boldsymbol{D}}^{(m)} \boldsymbol{z}^{(m)}$$

$$+\left(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right)^{\mathrm{T}} \underline{\boldsymbol{D}}^{(m)} \mathrm{diag}\left(\underline{\boldsymbol{E}}^{(m)}\right) + \boldsymbol{q}^{(m)^{\mathrm{T}}}\widetilde{\boldsymbol{\beta}}^{(m+1)} + c^{(m)} \qquad \text{(B11)}$$

The sign of each multiplier term follows the split-neuron convention defined in (21) of the main paper. The added Lagrangian term preserves a valid lower bound for either the positive or negative split.

Now, we add the multiplier $\boldsymbol{\beta}^{(m)}$ to $\boldsymbol{z}^{(m)}$ to enforce per-neuron constraints:

$$\min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} f(\boldsymbol{x}) \geq \max_{\widetilde{\boldsymbol{\beta}}^{(m+1)}\geq 0} \quad \min_{\boldsymbol{x}\in\mathcal{B},\tilde{\boldsymbol{z}}^{(m-1)}\in\tilde{\mathcal{U}}^{(m-1)}} \quad \max_{\boldsymbol{\beta}^{(m)}\geq 0} \quad \left(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right)^{\mathrm{T}} \underline{\boldsymbol{D}}^{(m)} \boldsymbol{z}^{(m)}$$

$$+\left(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right)^{\mathrm{T}} \underline{\boldsymbol{D}}^{(m)} \mathrm{diag}\left(\underline{\boldsymbol{E}}^{(m)}\right) + \boldsymbol{q}^{(m)^{\mathrm{T}}}\widetilde{\boldsymbol{\beta}}^{(m+1)} + c^{(m)}$$

$$\geq \max_{\widetilde{\boldsymbol{\beta}}^{(m)}\geq 0} \quad \min_{\boldsymbol{x}\in\mathcal{B},\tilde{\boldsymbol{z}}^{(m-1)}\in\tilde{\mathcal{U}}^{(m-1)}} \left(\boldsymbol{a}^{(m)^{\mathrm{T}}} \underline{\boldsymbol{D}}^{(m)} + \widetilde{\boldsymbol{\beta}}^{(m+1)^{\mathrm{T}}} \boldsymbol{P}^{(m)^{\mathrm{T}}} \underline{\boldsymbol{D}}^{(m)} + \boldsymbol{\beta}^{(m)^{\mathrm{T}}} \boldsymbol{K}^{(m)}\right) \boldsymbol{z}^{(m)}$$

$$+\left(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\right)^{\mathrm{T}} \underline{\boldsymbol{D}}^{(m)} \mathrm{diag}\left(\underline{\boldsymbol{E}}^{(m)}\right) + \boldsymbol{q}^{(m)^{\mathrm{T}}}\widetilde{\boldsymbol{\beta}}^{(m+1)} + c^{(m)} \qquad \text{(B12)}$$

By Applying weak duality, the $\max - \min$ expression provides a lower bound on the corresponding $\min - \max$ problem, ensuring that each maximization over $\boldsymbol{\beta}^{(i)}$ precedes the minimization (see (22) and (23) of the main paper). Then, we substitute $\boldsymbol{z}^{(m)} = \boldsymbol{W}^{(m)}\boldsymbol{z}^{(m-1)} + \boldsymbol{b}^{(m)}$ and obtain:

$$\min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} f(\boldsymbol{x}) \geq \max_{\widetilde{\boldsymbol{\beta}}^{(m)}\geq 0} \min_{\boldsymbol{x}\in\mathcal{B},\tilde{\boldsymbol{z}}^{(m-1)}\in\tilde{\mathcal{U}}^{(m-1)}} \Big(\boldsymbol{a}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)} + \widetilde{\boldsymbol{\beta}}^{(m+1)\mathrm{T}}\boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}$$

$$+ \boldsymbol{\beta}^{(m)\mathrm{T}}\boldsymbol{K}^{(m)}\Big)\boldsymbol{W}^{(m)}\hat{\boldsymbol{z}}^{(m-1)}$$

$$+ \Big(\boldsymbol{a}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)} + \widetilde{\boldsymbol{\beta}}^{(m+1)\mathrm{T}}\boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)} + \boldsymbol{\beta}^{(m)\mathrm{T}}\boldsymbol{K}^{(m)}\Big)^{\mathrm{T}}\boldsymbol{b}^{(m)}$$

$$+ \big(\boldsymbol{a}^{(m)} + \boldsymbol{P}^{(m)}\widetilde{\boldsymbol{\beta}}^{(m+1)}\big)^{\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\big(\underline{\boldsymbol{E}}^{(m)}\big) + \boldsymbol{q}^{(m)\mathrm{T}}\widetilde{\boldsymbol{\beta}}^{(m+1)} + c^{(m)}$$

$$= \Big[\underbrace{\Big[\boldsymbol{a}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{W}^{(m)}\Big]^{\mathrm{T}}}_{\boldsymbol{a}'} + \underbrace{(\widetilde{\boldsymbol{\beta}}^{(m+1)\mathrm{T}}\boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{W}^{(m)} + \boldsymbol{\beta}^{(m)\mathrm{T}}\boldsymbol{K}^{(m)}\boldsymbol{W}^{(m)})^{\boldsymbol{T}}}_{\boldsymbol{P}'\widetilde{\boldsymbol{\beta}}^{(m)}}\Big]^{\mathrm{T}}\hat{\boldsymbol{z}}^{(m-1)}$$

$$+\underbrace{\Big(\Big(\boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{b}^{(m)} + \boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(m)}) + \boldsymbol{q}^{(m)}\Big)^{\mathrm{T}}\widetilde{\boldsymbol{\beta}}^{(m+1)} + \big(\boldsymbol{K}^{(m)}\boldsymbol{b}^{(m)}\big)^{\mathrm{T}}\boldsymbol{\beta}^{(m)}\Big)}_{\boldsymbol{q}'^{\mathrm{T}}\widetilde{\boldsymbol{\beta}}^{(m)}}$$

$$+\underbrace{\boldsymbol{a}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{b}^{(m)} + \boldsymbol{a}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\big(\underline{\boldsymbol{E}}^{(m)}\big) + c^{(m)}}_{c'} \tag{B13}$$

Now, we evaluate each term $\boldsymbol{a}', \boldsymbol{P}', \boldsymbol{q}'$ and c' to demonstrate that the induction holds. For $\boldsymbol{a}'$ and c' we have:

$$\boldsymbol{a}' = \Big[\boldsymbol{a}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{W}^{(m)}\Big]^{\mathrm{T}} = \big[\Psi(L,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{W}^{(m)}\big]^{\mathrm{T}} = \big[\Psi(L,m)\boldsymbol{W}^{(m)}\big]^{\mathrm{T}}$$

$$= \boldsymbol{a}^{(m-1)} \tag{B14}$$

$$c' = c^{(m)} + \Psi(L,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{b}^{(m)} + \Psi(L,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\big(\underline{\boldsymbol{E}}^{(m)}\big)$$

$$= \sum_{i=m+1}^{L}\Psi(L,i)\boldsymbol{b}^{(i)} + \sum_{i=m+2}^{L}\Psi(L,i)\boldsymbol{W}^{(i)}\underline{\boldsymbol{D}}^{(i-1)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(i-1)}) + \Psi(L,m)\boldsymbol{b}^{(m)}$$

$$+\Psi(L,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\big(\underline{\boldsymbol{E}}^{(m)}\big)$$

$$= \sum_{i=m}^{L}\Psi(L,i)\boldsymbol{b}^{(i)} + \sum_{i=m+1}^{L}\Psi(L,i)\boldsymbol{W}^{(i)}\underline{\boldsymbol{D}}^{(i-1)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(i-1)})$$

$$= c^{(m-1)} \tag{B15}$$

For $\boldsymbol{P}' \coloneqq \left[\boldsymbol{P}'_m{}^{\mathrm{T}} \ldots \boldsymbol{P}'_{L-1}{}^{\mathrm{T}}\right]$, we have a new block of $\boldsymbol{P}'_m$where,

$$\boldsymbol{P}'_m = \boldsymbol{K}^{(m)}\boldsymbol{W}^{(m)} = \boldsymbol{K}^{(m)}\Psi(m,m)\boldsymbol{W}^{(m)} = \boldsymbol{P}_m^{(m-1)} \tag{B16}$$

For other blocks where $m+1 \le i \le L-1$,

$$\boldsymbol{P}'_i = \boldsymbol{P}_i^{(m)}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{W}^{(m)} = \boldsymbol{K}^{(i)}\Psi(i,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{W}^{(m)} = \boldsymbol{K}^{(i)}\Psi(i,m)\boldsymbol{W}^{(m)}$$

$$= \boldsymbol{P}_i^{(m-1)} \tag{B17}$$

For $\boldsymbol{q}' \coloneqq \left[\boldsymbol{q}'_m{}^{\mathrm{T}} \ldots \boldsymbol{q}'_{L-1}{}^{\mathrm{T}}\right]$, we have a new block of $\boldsymbol{q}'_m$where,

$$\boldsymbol{q}'_m = \boldsymbol{K}^{(m)}\boldsymbol{b}^{(m)} = \sum_{k=m}^{m} \boldsymbol{K}^{(m)}\Psi(m,k)\boldsymbol{b}^{(k)} = \boldsymbol{q}_m^{(m-1)} \tag{B18}$$

For other blocks where $m+1 \le i \le L-1$ we have,

$$\boldsymbol{q}'_i = \boldsymbol{q}_i^{(m)} + \boldsymbol{P}^{(m)}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{b}^{(m)} + \boldsymbol{P}^{(m)}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\left(\underline{\boldsymbol{E}}^{(m)}\right)$$

$$= \sum_{k=m+1}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{b}^{(k)} + \sum_{k=m+2}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{W}^{(k)}\underline{\boldsymbol{D}}^{(k-1)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(k-1)})$$

$$+\boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{b}^{(m)} + \boldsymbol{P}^{(m)\mathrm{T}}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\left(\underline{\boldsymbol{E}}^{(m)}\right)$$

$$= \sum_{k=m+1}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{b}^{(k)} + \sum_{k=m+2}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{W}^{(k)}\underline{\boldsymbol{D}}^{(k-1)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(k-1)})$$

$$+\boldsymbol{K}^{(i)}\Psi(i,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\boldsymbol{b}^{(m)} + \boldsymbol{K}^{(i)}\Psi(i,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\left(\underline{\boldsymbol{E}}^{m}\right)$$

$$= \sum_{k=m+1}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{b}^{(k)} + \sum_{k=m+2}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{W}^{(k)}\underline{\boldsymbol{D}}^{(k-1)}\underline{\boldsymbol{E}}^{(k-1)} + \boldsymbol{K}^{(i)}\Psi(i,m)\boldsymbol{b}^{(m)}$$

$$+\boldsymbol{K}^{(i)}\Psi(i,m+1)\boldsymbol{W}^{(m+1)}\underline{\boldsymbol{D}}^{(m)}\mathrm{diag}\left(\underline{\boldsymbol{E}}^{(m)}\right)$$

$$= \sum_{k=m}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{b}^{(k)} + \sum_{k=m+1}^{i} \boldsymbol{K}^{(i)}\Psi(i,k)\boldsymbol{W}^{(k)}\underline{\boldsymbol{D}}^{(k-1)}\mathrm{diag}(\underline{\boldsymbol{E}}^{(k-1)})$$

$$= \boldsymbol{q}_i^{(m-1)} \tag{B19}$$

Thus, $\boldsymbol{a}' = \boldsymbol{a}^{(m-1)}, \boldsymbol{P}' = \boldsymbol{P}^{(m-1)}, \boldsymbol{q}' = \boldsymbol{q}^{(m-1)}$ and $c' = c^{(m-1)}$ so the induction holds for layer $\hat{\boldsymbol{z}}^{(m-1)}$:

$$\min_{\boldsymbol{x}\in\mathcal{B},\boldsymbol{z}\in\mathcal{U}} f(\boldsymbol{x}) \geq \max_{\widetilde{\boldsymbol{\beta}}^{(m)}\geq 0} \min_{\boldsymbol{x}\in\mathcal{B},\tilde{\boldsymbol{z}}^{(m-1)}\in\tilde{\mathcal{U}}^{(m-1)}} \left(\boldsymbol{a}^{(m-1)} + \boldsymbol{P}^{(m-1)}\widetilde{\boldsymbol{\beta}}^{(m)}\right)^{\mathrm{T}} \hat{\boldsymbol{z}}^{(m-1)} + \boldsymbol{q}^{(m-1)^{\mathrm{T}}} \widetilde{\boldsymbol{\beta}}^{(m)} + c^{(m-1)} \tag{B20}$$

Therefore, the result holds for all layers. Finally, setting k to the value specified in (B1)–(B4) yields the bound stated in Theorem 2, completing the proof.

■

REFERENCE

[B1] S. Wang *et al*., "Beta-CROWN: efficient bound propagation with per-neuron split constraints for neural network robustness verification," in *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp. 29909–29921.